Ollivier–Ricci Curvature as a Landscape Analysis Framework for Discrete Combinatorial Optimization

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Abstract. We introduce the *ORC Transition Graph* (OTG), a deterministic directed graph over local optima derived from Ollivier–Ricci curvature (ORC), as a structural analysis tool for discrete fitness landscapes. Each local optimum has exactly one outgoing edge to the optimum reached by following its most-negative-curvature direction and hill-climbing—a purely local, $O(k^2)$ computation. We study OTG on NK landscapes, epistatic W-model benchmarks ($N = 16$, full enumeration), random MAX-SAT (N up to 100), QAP (permutation swaps, n up to 9), and TSP (2-opt, n up to 50 via sampling). The OTG extracts cross-basin information invisible to standard local optima networks (LONs): using only 1-hop neighborhood data, it achieves up to 9.2× better attractor quality than LON-d1 at equal information cost on epistatic W-model landscapes, and even outperforms the stochastic 3-hop LON-d3 on those instances. A structural crossover in OTG topology—from acyclic to cyclic at moderate epistasis—correlates with when ORC-guided perturbation outperforms random perturbation in iterated local search (ILS). The OTG framework applies to any graph with the disjoint-neighborhood property (Proposition 1), which we prove covers binary hypercubes, permutation swap graphs, *and* TSP 2-opt neighborhoods—the three foundational neighborhoods in combinatorial optimization. ORC's per-step escape advantage grows monotonically with neighborhood size—from 3.1× to 18.3× over random on TSP ($n = 8$ to 50)—though its computational cost ($O(k^2)$ per optimum) limits algorithmic use to settings where fitness evaluations are expensive. Per-optimum ORC metrics converge from ∼20 sampled optima (<1% error), while full OTG structural analysis remains limited to enumerable landscapes.

Keywords: Fitness landscape analysis · Ollivier–Ricci curvature · Local optima networks · Discrete optimization · Permutation spaces

1 Introduction

Fitness landscape analysis (FLA) provides the theoretical foundation for understanding why metaheuristic optimizers succeed or fail on specific problem

instances [16] and for automated algorithm selection [4]. Classical metrics—fitness-distance correlation (FDC) [2], autocorrelation length [19], information content [23], and exploratory landscape analysis (ELA) features [8]—characterize global statistical properties such as ruggedness and smoothness. Local optima networks (LONs) [12,14] go further by representing basin connectivity as a graph whose edges are discovered through stochastic perturbation followed by hill climbing (HC).

Both approaches share a limitation: classical metrics provide scalar summaries blind to local geometric structure, while LONs provide connectivity but not *directional* information—at a local optimum, which specific neighbor escapes the basin? LON construction also depends on perturbation distance d and requires many stochastic trials per optimum.

We address these limitations through Ollivier–Ricci curvature (ORC) [15], a discrete analogue of Ricci curvature that measures how neighborhood distributions of adjacent nodes diverge or converge. Negative curvature identifies directions where the landscape geometry "spreads apart"—the signature of inter-basin saddle regions. From ORC, we construct the *ORC Transition Graph* (OTG): a deterministic directed graph where each local optimum points to the optimum reached by following its min-ORC direction.

Contributions.

1. We formalize fitness-lifted ORC for discrete optimization landscapes and prove that negative curvature is guaranteed at every local optimum on graphs with the disjoint-neighborhood property, covering binary hypercubes, permutation swap graphs, and TSP 2-opt neighborhoods.
2. We introduce the OTG, a deterministic, parameter-free directed graph over local optima derived from optimal-transport-based curvature. On epistatic binary landscapes, the OTG achieves up to 9.2× better attractor quality than LON-d1 at equal information cost; across binary, permutation, and tour search spaces, ORC's per-step escape advantage grows monotonically with neighborhood size.
3. We validate the framework algorithmically through ORC-guided iterated local search (ILS) and show that OTG structural properties—specifically the presence or absence of cycles—indicate when ORC-guided perturbation adds value over random perturbation.

2 Background

A *fitness landscape* $(S, \mathcal{N}, f)$ consists of solutions S, neighborhood function $\mathcal{N} : S \to 2^S$, and fitness $f : S \to \mathbb{R}$ [19]. A solution x^* is a *local optimum* (minimization) if $f(x^*) \leq f(y)$ for all $y \in \mathcal{N}(x^*)$. LONs [12,13] compress the landscape into a graph whose nodes are local optima and whose edges encode transitions discovered by perturbing a solution (flipping d random bits) and hill-climbing. LON features—including funnel structure [11], fractal dimension [20],

and phase-transition indicators [10]—predict search difficulty [14], but edge construction requires T stochastic trials per optimum (typically $T \geq 50$) and the resulting funnel decomposition depends on d.

Ollivier [15] defined Ricci curvature for metric-measure spaces; Lin et al. [6] gave a modified graph formulation. For a graph edge (x, y):

$$\kappa(x, y) = 1 - \frac{W_1(\mu_x, \mu_y)}{d(x, y)} \tag{1}$$

where μ_x, μ_y are neighborhood probability measures and W_1 is the Wasserstein-1 distance [24]. Negative curvature signals diverging neighborhoods—a signature of community bridges [9] and GNN bottlenecks [22,18]; positive curvature signals contraction. We choose Ollivier's formulation over Forman's [17]: on triangle-free graphs (including all three neighborhoods in Corollary 1), Forman curvature is the constant $4-2k$ for every edge on unweighted k-regular graphs, and depends only on endpoint-local quantities even with fitness weighting—missing the distributional matching that gives ORC its cross-basin signal. To our knowledge, ORC has not been applied to fitness landscape analysis.

3 ORC on Fitness Landscapes

3.1 Fitness-Lifted ORC

We define ORC on the search graph $(S, \mathcal{N})$ with uniform measures μ_x over $\{x\} \cup \mathcal{N}(x)$. The transport cost between $a \in \text{supp}(\mu_x)$ and $b \in \text{supp}(\mu_y)$ is:

$$c(a, b) = \delta(a, b) + \gamma \cdot |f(a) - f(b)| \tag{2}$$

where $\gamma \geq 0$ controls fitness sensitivity and the overlap indicator is:

$$\delta(a, b) = \begin{cases} 0 & \text{if } a = b \\ 1 & \text{if } a \text{ or } b \in \text{supp}(\mu_x) \cap \text{supp}(\mu_y) \\ 2 & \text{otherwise (both exclusive)} \end{cases} \tag{3}$$

This is a conservative coarse-graining of graph distance computable from 2-hop information alone, preserving $O(k^2)$ locality. For uniform measures of equal support size, W_1 reduces to an optimal assignment problem solved by the Hungarian algorithm in $O(k^3)$ [5]; the fine-grained complexity of this reduction has been formally established [1].

Parameter sensitivity. We use $\gamma = 1$ throughout. Importantly, the ORC-guided *direction* (Algorithm 1) is γ-invariant: changing γ scales all fitness differences uniformly, so the optimal transport matching order—and thus the min-ORC neighbor—does not change. Empirically, escape rates are identical to three decimal places across $\gamma \in \{0.1, 0.5, 1.0, 2.0\}$ on all 480 instances; only absolute curvature values change. The framework is thus parameter-free for navigation; γ matters only for difficulty prediction, where $\gamma = 1$ balances structural and fitness terms. This invariance is a consequence of the *linear* fitness term in Eq. (2): a nonlinear cost $g(|f(a) - f(b)|)$ would break it, though such extensions are a natural direction for future work.

3.2 Why ORC Sees What 1-Hop Methods Cannot

Proposition 1 (ORC Cross-Neighborhood Decomposition). *Let $G = (V, E)$ be a regular graph of degree k in which adjacent nodes share no common neighbors: $\mathcal{N}(x) \cap \mathcal{N}(y) = \emptyset$ for all $(x, y) \in E$. Then $|\mathrm{supp}(\mu_{x^*}) \cap \mathrm{supp}(\mu_y)| = 2$ (only x^* and y are shared), and the optimal transport cost decomposes as:*

$$(k{+}1)\, W_1(\mu_{x^*}, \mu_y) = C_{\text{shared}} + \min_{\sigma} \sum_{i=1}^{k-1} \big[2 + \gamma\, |f(a_i) - f(b_{\sigma(i)})|\big] \tag{4}$$

where $C_{\text{shared}} \leq 2 + \gamma(\Delta + \Delta')$ accounts for the two shared elements, $\{a_i\} = \mathcal{N}(x^*) \setminus \{y\}$, $\{b_j\} = \mathcal{N}(y) \setminus \{x^*\}$ are the $k{-}1$ exclusive neighbors on each side, and σ ranges over permutations of $\{1, \ldots, k{-}1\}$.

Proof. Since $\mathcal{N}(x^*) \cap \mathcal{N}(y) = \emptyset$, the only shared support elements are x^* and y themselves. Any assignment $\pi : \mathrm{supp}(\mu_{x^*}) \to \mathrm{supp}(\mu_y)$ partitions into: (i) the two shared positions (x^*, y) contributing C_{shared}, and (ii) the k–1 exclusive pairs, each incurring $\delta = 2$ plus a fitness-difference term. The optimal σ minimizes the total fitness-comparison cost.

Corollary 1 (Applicability to Standard Search Spaces). *Proposition 1 applies to: **(i)** the N-dimensional binary hypercube with bit-flip neighborhood ($k = N$), **(ii)** the symmetric group S_n with pairwise-swap neighborhood ($k = \binom{n}{2}$), and **(iii)** Hamiltonian tours with 2-opt neighborhood ($k = n(n{-}3)/2$).*

Proof. **(i)** On the hypercube, flipping bit i in x and bit j in $y = x \oplus e_p$ yields the same neighbor only if $e_i = e_p \oplus e_j$; since single bit-flips cannot equal a two-bit change, $\mathcal{N}(x) \cap \mathcal{N}(y) = \emptyset$. **(ii)** On S_n, if $y = x \circ \tau_{ab}$, a common neighbor z requires $\tau_{ij} = \tau_{ab} \circ \tau_{kl}$ for transpositions τ_{ij}, τ_{kl}. The product of two distinct transpositions is a 3-cycle or double transposition—never a transposition. **(iii)** A 2-opt move is a segment reversal $\rho_{p,q}$ mapping position $i \mapsto p{+}q{+}1{-}i$ for $i \in \{p{+}1, \ldots, q\}$ and fixing all others—an affine map with *slope* -1 on its support. A common neighbor requires $\rho_{a,b} = \rho_{c,d} \circ \rho_{p,q}$ for distinct non-trivial reversals. *Disjoint supports:* the composition has non-contiguous support (two separated intervals with a fixed gap), impossible for a single reversal. *One nested in the other* (WLOG $\mathrm{supp}(\rho_{p,q}) \subset \mathrm{supp}(\rho_{c,d})$): on the inner interval, $(\rho_{c,d} \circ \rho_{p,q})(i) = i + \delta$ is a *translation* (slope $+1$), since each reversal contributes slope -1 and $(-1)(-1) = +1$. Any single reversal has slope -1 on its entire support, so the composition cannot be one. *Partial overlap:* within the overlap the composition transitions from slope $+1$ (where both act) to slope -1 (where only one acts)—again incompatible with a single reversal. Verified computationally at $n \leq 8$ (all 25,200 edges) and by sampling at $n \leq 11$ (all >200,000 tested edges have overlap 0).

These three search spaces—binary strings (SAT, NK), permutations with swaps (QAP, scheduling), and tours with 2-opt (TSP)—cover the foundational neighborhoods in combinatorial optimization.

Algorithm 1 ORC-Guided Escape from a Local Optimum

Require: Local optimum x^*, fitness f, neighborhood $\mathcal{N}$, parameter γ
1: **for** each $y \in \mathcal{N}(x^*)$ **do**
2: Compute $\kappa(x^*, y)$ via Eq. (1)–(3)
3: **end for**
4: $y^* \leftarrow \arg\min_y \kappa(x^*, y)$
5: **return** HillClimb(y^*)

The exclusive-pair term is the key: it computes an optimal matching between the fitness distributions visible from x^* and from y. At a basin boundary, the two sides' fitness distributions diverge, making curvature strongly negative. By contrast, LON-d1 evaluates each neighbor y using only $f(y)$ and the downstream hill-climbing destination, discarding the $(k-1)$ exclusive neighbors entirely—precisely the distributional comparison that ORC captures.

3.3 Negative Curvature Guarantee

Theorem 1. *Let x^* be a strict local optimum on any degree-k graph satisfying the disjoint-neighborhood condition of Proposition 1 ($k \geq 4$), with cost* (2)–(3) *and $\gamma > 0$. If neighbor y satisfies $\Delta = f(y) - f(x^*) < (k-3)/((k+1)\gamma)$, then $\kappa(x^*, y) < 0$.*

Proof. From Proposition 1, every assignment incurs exclusive-pair structural cost at least $2(k-1)$, giving $W_1 \geq 2(k-1)/(k+1)$. With edge cost $c(x^*, y) = 1 + \gamma\Delta$: $\kappa(x^*, y) \leq 1 - 2(k-1)/[(k+1)(1+\gamma\Delta)]$, which is negative iff $\Delta < (k-3)/((k+1)\gamma)$.

For the binary hypercube with $k = N = 16$, $\gamma = 1$: the bound requires $\Delta < 0.76$. For QAP with $n = 8$ ($k = 28$): $\Delta < 0.86$. Across all 5,000+ local optima in our binary benchmarks and 1,500+ QAP optima, **100% satisfy the condition**. The min-ORC neighbor $y^* = \arg\min_y \kappa(x^*, y)$ identifies the direction of maximal neighborhood divergence—the strongest saddle direction (Alg. 1).

3.4 ORC-Guided Escape and the OTG

Applying Algorithm 1 at every local optimum of a landscape defines the *ORC Transition Graph* (OTG): a deterministic directed graph where each optimum has exactly one outgoing edge to the optimum reached by following its min-ORC neighbor and hill-climbing. The OTG is a functional graph; its terminal attractors are either self-loops (sinks) or directed cycles. The *funnel* of an attractor is the set of all optima whose OTG path converges to it. Unlike LONs [21], the OTG is deterministic and parameter-free: it requires no perturbation distance or stochastic trials.

4 Experimental Setup

4.1 Benchmark Problems

We evaluate on five benchmark families spanning three search spaces:

NK landscapes [3] define tunably rugged fitness via K epistatic interactions per bit on length-N binary strings. We test $K \in \{4, 8, 12\}$ at $N = 16$ (30 instances each, full enumeration).

W-model [25] applies neutrality, epistasis, and ruggedness transformations controlled by a difficulty parameter ν to a base OneMax problem. We test $\nu \in \{3, 4, 6\}$ at $N = 16$ (30 instances each, full enumeration).

Random MAX-SAT. Random 3-SAT instances with $N \in \{20, 50, 100\}$ variables and clause-to-variable ratio $\alpha \in \{3.0, \ldots, 6.0\}$ (20 instances each). Local optima are collected via random-restart HC (500–2,000 restarts). This domain has well-studied phase transitions [10] and no structural overlap with NK or W-model.

Random QAP. Quadratic assignment problems with uniform random flow/distance matrices and pairwise-swap neighborhood, $n \in \{6, 7, 8, 9\}$ (20 instances each, full enumeration).

Random TSP. Random-matrix TSP instances with 2-opt neighborhood. Full enumeration at $n \in \{8, 9, 10\}$ (20 each, 4–7 optima per instance); sampling-based analysis at $n \in \{15, 20, 30, 50\}$ (20 each, 300 random-restart 2-opt runs per instance). TSP instances at small n serve to validate the theoretical framework on an exact ground truth; sampling-based runs at larger n test scaling.

The five families were chosen to cover three distinct search spaces (binary strings, permutations, tours), a range of landscape ruggedness levels, and both enumerable and sampling-only regimes.

4.2 Algorithms and Baselines

Hill Climbing (HC). Best-improvement (steepest descent) HC: at each step, all k neighbors are evaluated and the one with the best fitness is selected. HC terminates when no neighbor improves on the current solution, yielding a local optimum. Ties are broken uniformly at random.

Random-Restart Hill Climbing (RR-HC). Repeat HC from uniformly random initial solutions until the budget of fitness evaluations (FEs) is exhausted. Return the best local optimum found.

Iterated Local Search (ILS) [7]. From a local optimum, apply a perturbation (bit-flips) and re-run HC; repeat until the FE budget is exhausted. We test four ILS variants, all using a fixed perturbation strength of 3 bit-flips for fair comparison:

- **Rand-ILS**: flip 3 random bits.
- **ORC+Pert**: flip the min-ORC bit (Algorithm 1) plus 2 random bits.
- **ORC-ILS**: flip only the min-ORC bit (1 bit; no random component).
- **MG-ILS**: flip only the MinGap bit (1 bit; see below).

Table 1. ORC escape direction analysis on binary landscapes ($N = 16$, full enumeration). %ORC: fraction of local optima where Algorithm 1 reaches a better optimum. %MinGap: same using the argmin-$|\Delta|$ heuristic. %Rand: random neighbor baseline (30 trials). The ORC advantage over MinGap emerges on epistatic W-model landscapes where the fitness-distance relationship is distorted.

Config	#Opt	%ORC	%MinGap	%Rand	ORC/R	ORC/MG	Agree
NK $K=4$	170	59.6	57.4	31.6	1.9×	1.04×	89.6%
NK $K=8$	888	58.5	56.5	33.6	1.7×	1.04×	90.3%
NK $K=12$	2249	51.1	49.4	37.6	1.4×	1.03×	91.7%
W $\nu=3$	406	79.9	43.5	29.4	**2.7**×	**1.84**×	59.6%
W $\nu=4$	3732	59.7	25.9	39.5	1.5×	**2.31**×	43.3%
W $\nu=6$	5836	63.9	37.4	48.6	1.3×	1.71×	52.7%

ORC-ILS and MG-ILS use only 1-bit perturbation; they are included as ablations to isolate the effect of the directed bit.

MinGap. The neighbor y with the smallest fitness difference $|f(y) - f(x^*)|$ (argmin-$|\Delta|$ heuristic). This greedy baseline uses only fitness values, not distributional information.

LON-d1. At a local optimum x^*, the LON-d1 destination is the mode HC outcome over all k single-bit-flip neighbors (1-hop, deterministic).

LON-d3. At a local optimum x^*, the LON-d3 destination is the mode HC outcome from 100 random 3-bit perturbations (3-hop, stochastic).

4.3 Evaluation Metrics

Escape rate (%). Fraction of local optima where a method reaches a strictly better optimum after one perturbation + HC step. This is the primary metric shared across all experiments.

Mean terminal rank. Normalized fitness rank of the attractor reached by following OTG/LON paths (0.0 = best, 0.5 = random).

Top-5% attractor (%). Fraction of optima whose OTG/LON path terminates at an attractor in the top 5% by fitness.

Compression. #funnels / #optima—lower means better funnel aggregation.

Success rate (%). Fraction of ILS runs finding the global optimum within the FE budget (used for the ILS comparison in Section 5.4).

5 Results

5.1 Single-Step Escape Analysis

Table 1 presents the single-step escape analysis. On NK landscapes, ORC beats random by 1.4–1.9× but only marginally outperforms MinGap (1.03–1.04×); on

Table 2. OTG vs. LON funnel quality ($N = 16$, full enumeration). LON-d1: 1-hop, deterministic. LON-d3: 3-hop, 100 stochastic trials. Rank: 0.0 = best, 0.5 = random. Compression: #funnels / #optima.

		Mean terminal rank			Top-5% attractor			Compression		
Config	**#Opt**	**OTG**	**d1**	**d3**	**OTG**	**d1**	**d3**	**OTG**	**d1**	**d3**
NK $K=4$	170	.255	.490	**.181**	24.6	5.6	**30.1**	40	98	35
NK $K=8$	888	.231	.476	**.099**	24.0	6.0	**46.1**	42	97	16
W $\nu=3$	406	**.037**	.340	.077	**90.7**	23.2	77.3	15	98	21
W $\nu=4$	3732	.073	.258	**.012**	73.7	20.0	**97.8**	13	75	2
W $\nu=6$	5836	.066	.221	**.011**	68.5	18.3	**96.3**	14	68	1

these regular Hamming graphs the fitness gradient already captures most of the cross-basin signal. On W-model landscapes, where epistasis distorts the fitness-distance relationship, ORC's advantage over MinGap becomes substantial: 2.31× at $\nu=4$, with only 43% agreement between the two methods.

When ORC and MinGap disagree, ORC wins. On W $\nu = 3$, ORC's choice leads to a better basin in **78.7%** of disagreements vs. MinGap's 0.3% (21% ties). On NK $K=4,8$, ORC wins 43–44% vs. MinGap's 24–25%. MinGap almost never outperforms ORC when the two disagree, confirming that ORC's optimal transport matching captures cross-basin structure that the fitness-gap heuristic misses.

ORC ranks all escape directions. Spearman correlation between ORC values and destination quality: mean $\rho=0.44$ on W $\nu=3$ (95% positive), $\rho=0.33$ on NK—more negative curvature $\rightarrow$ better destination basin.

5.2 OTG vs. LON: Funnel Detection from Local Data

Table 2 compares OTG with LON variants. Both OTG and LON-d1 use only 1-hop data, making this an equal-information comparison. LON-d1 yields 97–98% singleton funnels; OTG compresses to 13–42% funnels with substantially better attractor quality. On W $\nu=3$: OTG rank 0.037 vs. LON-d1's 0.340 (9.2× improvement); 90.7% reach top-5% vs. 23.2%. The OTG even outperforms LON-d3 (3-hop, stochastic) on W $\nu = 3$ (rank 0.037 vs. 0.077); at higher epistasis ($\nu \geq 4$), LON-d3's wider perturbation provides superior attractor quality. Path to top-5% attractors: median 2 hops, P90 3 hops.

5.3 Scaling Across Problem Sizes and Search Spaces

MAX-SAT (binary strings). Table 3 summarizes results. Proposition 1's sort-and-match reduction lowers per-edge ORC cost from $O(k^3)$ to $O(k \log k)$, enabling computation at 500 optima for $N = 100$ in ∼18 s. The ORC/Random escape ratio increases from 2.4× at $N = 20$ to 3.7× at $N = 100$ ($\alpha = 4.27$; $\rho(N, \text{ORC/Rand}) = 0.77$, $p < 10^{-84}$). As N grows, the (N–1) exclusive-neighbor

Table 3. ORC escape advantage on random MAX-SAT. The ORC/Random ratio *increases* with N: ORC's distributional signal becomes more valuable in higher dimensions. 20 instances per cell; $\alpha_c \approx 4.27$ is the satisfiability phase transition.

N	α	%ORC	%Rand	%MG	ORC/R	ORC/MG	Mean ORC
20	3.00	34.3	13.5	26.1	2.5×	1.3×	−0.428
20	4.27	42.7	17.5	36.7	2.4×	1.2×	−0.374
20	6.00	50.0	20.1	46.3	2.5×	1.1×	−0.320
50	3.00	32.8	10.1	21.9	3.3×	1.5×	−0.493
50	4.27	37.3	10.6	31.3	3.5×	1.2×	−0.418
50	6.00	40.4	10.3	38.1	3.9×	1.1×	−0.353
100	3.00	34.0	9.9	21.4	3.4×	1.6×	−0.524
100	4.27	39.5	10.7	32.0	3.7×	1.2×	−0.450
100	6.00	42.7	10.1	40.2	4.2×	1.1×	−0.385

Table 4. OTG vs. LON-d1 on QAP (swap neighborhood, full enumeration). ORC's advantage grows with n. LON-d1 degrades to 100% singletons at $n = 9$. 20 instances per row.

				Escape rate		Mean rank		Compression	
n	k	#Opt	ORC/R	ORC	LON	OTG	LON	OTG	LON
6	15	12	1.8×	51.2	4.3	.230	.466	49%	96%
7	21	27	2.1×	55.9	3.0	.235	.476	44%	97%
8	28	78	2.3×	58.7	0.9	.230	.493	41%	99%
9	36	233	**2.7**×	65.0	0.1	**.194**	.499	35%	**100%**

comparisons carry more distributional information, strengthening the cross-basin signal. Mean ORC also tracks the SAT phase transition: $\rho(\text{ORC}, \alpha) = 0.94$ ($N = 20$), 0.99 ($N = 100$).

QAP (permutations). Table 4 shows that LON-d1 degrades to 0.1% escape at $n = 9$ (100% singletons, rank $\approx$ random); OTG improves: ORC/Random 1.8× → 2.7×, rank 0.194 with 35% compression. ORC's advantage over MinGap is modest (1.06×) on these random instances; the primary result is OTG vs. LON-d1.

TSP (2-opt). Table 5 confirms Corollary 1(iii): the disjoint-neighborhood property holds at all tested sizes. As k grows from 20 to 1175, random escape drops from 25% to 2.5% while ORC maintains 36–45%, yielding an advantage over Random that grows monotonically ($\rho(k, \text{ORC/Rand}) > 0.99$). ORC also consistently outperforms MinGap, winning 45–52% of disagreements vs. 9–22%. This pattern—growing ORC/Random advantage, stable ORC escape rate—is consistent across all three search spaces, confirming that ORC's distributional matching becomes more valuable as neighborhood size increases.

Table 5. ORC escape on TSP 2-opt across scales. Top rows: full enumeration. Bottom rows: sampling-based (300 restarts per instance). The ORC/Random ratio grows monotonically from 3.1× at $n=8$ to 18.3× at $n=50$. When ORC and MinGap disagree, ORC wins 45–52% vs. MinGap's 9–22%.

n	k	#Opt	Escape rate (%) ORC	MG	Rand	ORC/R	ORC/MG
Full enumeration							
8	20	4	77.5	66.7	25.0	3.1×	1.16×
9	27	5	64.7	57.5	21.1	3.1×	1.13×
10	35	7	68.9	62.6	21.3	3.2×	1.10×
Sampling-based (300 restarts)							
15	90	225	40.0	35.4	5.2	7.7×	1.13×
20	170	289	35.7	32.9	3.9	9.2×	1.09×
30	405	300	40.7	38.7	3.2	12.7×	1.05×
50	1175	300	45.0	43.5	2.5	18.3×	1.03×

Table 6. Fixed-budget ILS comparison at $N=16$ (5,000 FEs, 30 instances × 30 trials). ORC+Pert and Rand-ILS both flip 3 bits per perturbation; ORC-ILS and MG-ILS flip only 1 directed bit (ablations). Bold: best per row.

Config	ORC+Pert	Rand-ILS	RR-HC	ORC-ILS	MG-ILS
Success rate (% finding global optimum)					
NK $K=4$	77.3	**88.9**	88.4	6.7	6.1
NK $K=8$	42.0	**44.9**	45.2	1.3	1.3
W $\nu=3$	46.7	**80.2**	52.2	8.4	1.8
W $\nu=4$	**89.9**	69.3	39.7	1.6	1.0
W $\nu=6$	**87.3**	56.0	26.3	1.6	0.9

5.4 ORC-Guided Iterated Local Search

Table 6 compares ILS variants under a fixed 5,000 FE budget. ORC+Pert wins on 2 of 5 configurations and loses on 3, partitioning cleanly along OTG topology. In the *acyclic regime* (NK, W $\nu=3$), Rand-ILS dominates—sometimes by large margins (80.2% vs. 46.7% on W $\nu=3$)—because the landscape has clear funnel structure and ORC's directional constraint reduces exploration diversity. In the *cyclic regime* (W $\nu = 4,6$), ORC+Pert dominates: 89.9% vs. 69.3% at $\nu = 4$. ORC-ILS and MG-ILS, which flip only a single directed bit, perform poorly (1–8% success) because without random bits their perturbation deterministically revisits the same path and becomes trapped in OTG cycles.

Computational cost. At $N = 16$, ORC costs ∼120 FEs (∼3 ms); at $N = 100$, ∼5 000 FEs (∼18 ms). At $N = 100$ on MAX-SAT (10 s wall-clock budget), Rand-ILS completes 25× more iterations and finds better solutions; ORC's 3.7× per-step advantage cannot overcome the iteration deficit when evaluations cost

Table 7. Spearman correlations (ρ) between landscape metrics and algorithm performance ($N = 16$, 480 instances: 30 each for NK $K \in \{4, 8, 12, 15\}$ and W-model $\nu \in \{1, \ldots, 12\}$). ***: $p < 0.001$. Best per column in **bold**. †: requires full enumeration. HC: best-improvement hill climbing (success = global optimum found). EA: (μ+λ) evolutionary algorithm with μ=5, λ=10, bit-flip mutation (success rate). RS: random sampling baseline (mean fitness of 10^4 uniform samples).

Feature	Cost	HC succ.	EA succ.	RS mean
Mean ORC (ours)	$O(Rk^2)$	-0.670^{***}	$\mathbf{-0.652}^{***}$	-0.455^{***}
%ORC→better	$O(Rk^2)$	-0.435^{***}	-0.174^{***}	-0.615^{***}
FDC	$O(2^N)$	0.276^{***}	0.501^{***}	0.024
Autocorrelation	$O(L)$	0.543^{***}	0.544^{***}	0.372^{***}
Info. content H	$O(L)$	-0.308^{***}	0.306^{***}	-0.585^{***}
Mean $\lvert\Delta f\rvert$ (ELA)	$O(L)$	$\mathbf{-0.695}^{***}$	-0.083	$\mathbf{-0.769}^{***}$
NBC mean (ELA)	$O(L)$	-0.419^{***}	0.072	-0.487^{***}
Basin entropy†	$O(2^N)$	-0.731^{***}	-0.460^{***}	-0.610^{***}
OTG attractor rank†	$O(2^N)$	0.636^{***}	0.157	0.754^{***}
OTG compression†	$O(2^N)$	0.589^{***}	-0.094	0.748^{***}

$\sim$0.001 ms. ORC-guided search becomes cost-effective when individual evaluations cost ≥ 1 ms (e.g., simulation-based optimization).

6 Discussion

6.1 OTG Structural Properties

Figure 1 reveals a structural crossover: at W $\nu = 3$ the OTG is acyclic (91% top-5%); at ν=4, cycles emerge trapping 21% of optima (top-5% drops to 74%). NK landscapes show no cycles at any K. This crossover correlates with the ILS results in Table 6: in acyclic regimes, random perturbation suffices and ORC's directional constraint reduces diversity; in cyclic regimes, ORC+Pert's directed bit breaks cycles that trap random perturbation. We note this is an observed correlation on the tested benchmark families; validating it as a predictive criterion on unseen landscape types is an avenue for future work.

6.2 ORC as Difficulty Predictor

Table 7 evaluates ORC as a difficulty predictor alongside classical FLA metrics. Mean $|\Delta f|$ achieves the strongest individual correlation on HC ($\rho = -0.695$) and RS ($\rho = -0.769$), but fails on EA ($\rho = -0.083$, not significant). Information content and NBC show similar target-specificity, with sign reversals across algorithm types. Autocorrelation maintains consistent sign but with weaker magnitude on RS ($\rho = 0.372$). Mean ORC is the only tractable metric with strong correlations (all $|\rho| > 0.45$) of consistent sign across all three algorithm types, making it a

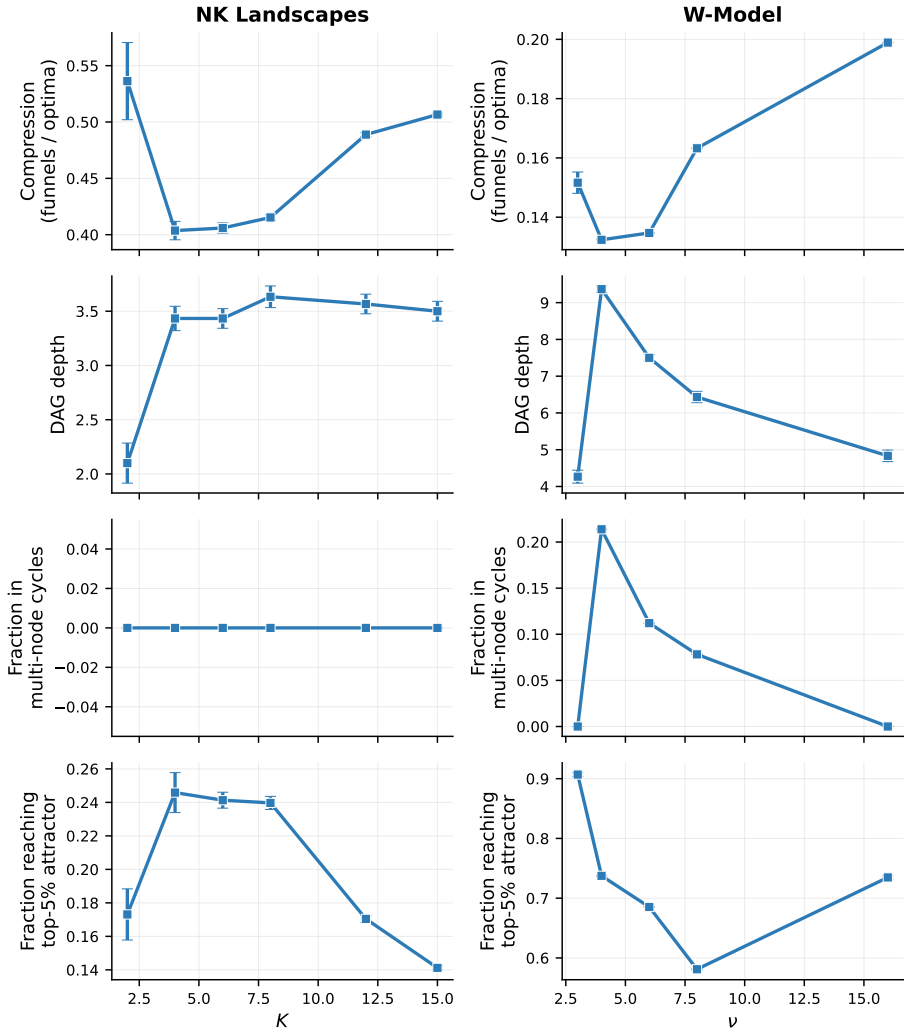

Fig. 1. OTG structural properties vs. ruggedness parameter. Left column: NK landscapes (K). Right column: W-model (ν). A structural crossover in cycle structure appears at W $\nu = 4$: multi-node cycles emerge (21% of optima), DAG depth peaks, and top-5% reachability drops. NK landscapes show no cycles at any K.

robust general-purpose predictor that does not require choosing the right metric for a specific algorithm family.

Table 8 extends the escape analysis to $N = 20$, confirming the pattern beyond the $N = 16$ regime: ORC's advantage decreases monotonically with K/N as maximal-ruggedness landscapes leave no exploitable distributional structure.

Table 8. ORC escape at $N=20$ (NK landscapes, full enumeration, 10 instances each). The ORC/Random advantage replicates the $N = 16$ pattern; at $K = 19$ ($\approx N$), all neighbors are equally rugged and ORC's advantage vanishes.

K	#Opt	%ORC	%Rand	ORC/R
4	557	62.3	31.2	2.0×
8	5,012	62.5	34.1	1.8×
15	26,086	52.9	37.2	1.4×
19	49,966	49.5	47.5	1.0×

6.3 OTG Visualization

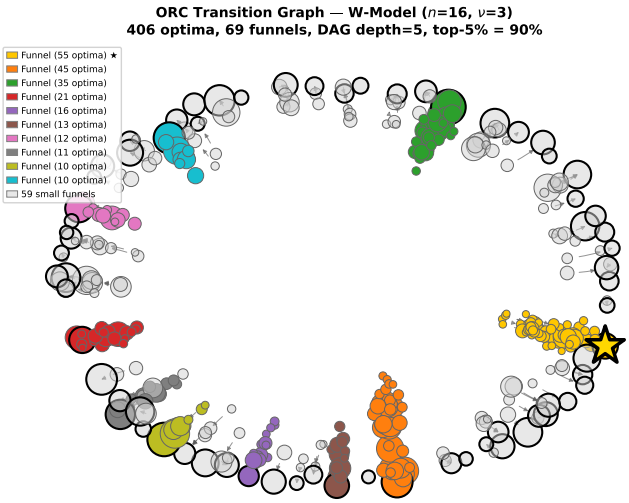

Fig. 2. OTG for a W-model instance ($N = 16$, $\nu = 3$, seed 0). Each node is a local optimum, sized by basin of attraction and colored by funnel membership. The gold star marks the global optimum. Edges follow the min-ORC direction deterministically. The 406 optima collapse into 69 funnels; 91% converge to a top-5% attractor. The circular layout groups funnels; spatial distance does not encode fitness or Hamming distance.

Figure 2 visualizes the OTG for a representative W-model instance. Clusters of optima flow deterministically toward high-quality attractors from local curvature alone, without stochastic trials or perturbation distance choices.

6.4 Convergence Under Sampling

On W-model at $N=16$, subsampling R optima uniformly (20 subsamples $\times$ 10 instances), per-optimum metrics converge fast: escape rate within 2% error at $R=5$, <1% at $R=20$. Structural metrics converge slowly—at 50% coverage, attractor rank is still 3.4$\times$ worse than ground truth, because the OTG cannot route to unsampled attractors. Per-optimum ORC metrics are thus reliable from $\sim$20 restarts at any scale; full OTG structural analysis remains limited to enumerable landscapes.

6.5 Limitations

The strongest OTG results (up to 9.2$\times$ attractor quality improvement over LON-d1) come from full enumeration at $N=16$ on W-model—a scale where exhaustive analysis is feasible but not representative of real-world problem sizes. At $N=100$ on MAX-SAT, Rand-ILS outperforms ORC+Pert in wall-clock time when evaluations are cheap ($\ll$1 ms). On QAP, ORC's advantage over MinGap is marginal (1.06$\times$) on random instances; structured QAPLIB instances may behave differently. For neighborhoods *without* the disjoint-neighborhood property (e.g., 3-opt, LK moves), Proposition 1's decomposition does not apply and ORC requires the full $O(k^3)$ Hungarian algorithm; characterizing ORC's behavior on such graphs is an open question. A full comparison with the flacco/ELA ecosystem [8] in an algorithm selection pipeline remains future work.

7 Conclusion

The ORC Transition Graph demonstrates that optimal transport over neighborhood fitness distributions captures cross-basin structure invisible to 1-hop methods, with the attractor quality gap over LON-d1 reaching up to 9.2$\times$ on epistatic W-model landscapes at $N = 16$. The disjoint-neighborhood property (Corollary 1) gives the framework a unified foundation across the three foundational neighborhoods in combinatorial optimization: bit-flips, permutation swaps, and TSP 2-opt. ORC's per-step escape advantage grows monotonically with neighborhood size (3.1$\times$ to 18.3$\times$ over random on TSP from $n=8$ to $n=50$; analogous trends on binary and swap spaces), though translating this into algorithmic gains remains cost-dependent. OTG structural properties—particularly the acyclic-to-cyclic crossover—correlate with when ORC-guided perturbation adds value in ILS, providing practitioners with a diagnostic signal. Per-optimum metrics converge from $\sim$20 samples, making ORC a practical addition to the FLA toolkit.

Reproducibility. All code and data are available at
`https://github.com/rhordoan/RiemannianSwarmSurgery`.

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References

1. DasGupta, B., Grigorescu, E., Mukherjee, T.: On computing discretized Ricci curvatures of graphs: Local algorithms and (localized) fine-grained reductions. arXiv:2208.09535 (2023)
2. Jones, T., Forrest, S.: Fitness distance correlation as a measure of problem difficulty for genetic algorithms. In: Eshelman, L.J. (ed.) 6th International Conference on Genetic Algorithms, pp. 184–192. Morgan Kaufmann, San Francisco (1995)
3. Kauffman, S.A.: The Origins of Order: Self-Organization and Selection in Evolution. Oxford University Press, New York (1993)
4. Kerschke, P., Hoos, H.H., Neumann, F., Trautmann, H.: Automated algorithm selection: Survey and perspectives. Evol. Comput. **27**(1), 3–45 (2019)
5. Kuhn, H.W.: The Hungarian method for the assignment problem. Naval Research Logistics **2**(1–2), 83–97 (1955)
6. Lin, Y., Lu, L., Yau, S.T.: Ricci curvature of graphs. Tohoku Math. J. **63**(4), 605–627 (2011)
7. Lourenço, H.R., Martin, O.C., Stützle, T.: Iterated local search: Framework and applications. In: Glover, F., Kochenberger, G.A. (eds.) Handbook of Metaheuristics, pp. 320–353. Springer, Boston (2003)
8. Mersmann, O., Bischl, B., Trautmann, H., Preuss, M., Weihs, C., Rudolph, G.: Exploratory landscape analysis. In: GECCO, pp. 829–836. ACM, New York (2011)
9. Ni, C.C., Lin, Y.Y., Luo, F., Gao, J.: Community detection on networks with Ricci flow. Sci. Rep. **9**, 9984 (2019)
10. Ochoa, G., Chicano, F., Tomassini, M.: Global landscape structure and the random MAX-SAT phase transition. In: Bäck, T., et al. (eds.) PPSN XVI. LNCS, vol. 12270, pp. 125–138. Springer, Cham (2020)
11. Ochoa, G., Herrmann, S., Rothlauf, F.: Communities of local optima as funnels in fitness landscapes. In: GECCO, pp. 325–331. ACM, New York (2016)
12. Ochoa, G., Tomassini, M., Vérel, S., Darabos, C.: A study of NK landscapes' basins and local optima networks. In: GECCO, pp. 555–562. ACM, New York (2008)
13. Ochoa, G., Veerapen, N.: Deconstructing the Big Valley search space hypothesis. In: Chicano, F., Hu, B., García-Sánchez, P. (eds.) EvoCOP. LNCS, vol. 9595, pp. 58–73. Springer, Cham (2016)
14. Ochoa, G., Veerapen, N., Daolio, F., Tomassini, M.: Understanding phase transitions with local optima networks: Number partitioning as a case study. In: Hu, B., López-Ibáñez, M. (eds.) EvoCOP. LNCS, vol. 10197, pp. 233–248. Springer, Cham (2017)
15. Ollivier, Y.: Ricci curvature of Markov chains on metric spaces. J. Funct. Anal. **256**(3), 810–864 (2009)
16. Pitzer, E., Affenzeller, M.: A comprehensive survey on fitness landscape analysis. In: Fodor, J., Klempous, R., Suárez Araujo, C.P. (eds.) Recent Advances in Intelligent Engineering Systems. SCI, vol. 378, pp. 161–191. Springer, Heidelberg (2012)
17. Samal, A., Sreejith, R.P., Gu, J., Liu, S., Saucan, E., Jost, J.: Comparative analysis of two discretizations of Ricci curvature for complex networks. Sci. Rep. **8**, 8650 (2018)
18. Sandhu, R., et al.: Graph curvature for differentiating cancer networks. Sci. Rep. **5**, 12323 (2015)
19. Stadler, P.F.: Landscapes and their correlation functions. J. Math. Chem. **20**, 1–45 (1996)

20. Thomson, S.L., Ochoa, G., Vérel, S.: The fractal geometry of fitness landscapes at the local optima level. Nat. Comput. **21**(2), 317–333 (2022)
21. Tomassini, M.: A local optima network view of real function fitness landscapes. Entropy **24**(5), 703 (2022)
22. Topping, J., Di Giovanni, F., Chamberlain, B.P., Dong, X., Bronstein, M.M.: Understanding over-squashing and bottlenecks on graphs via curvature. In: ICLR. OpenReview.net (2022)
23. Vassilev, V.K., Fogarty, T.C., Miller, J.F.: Information characteristics and the structure of landscapes. Evol. Comput. **8**(1), 31–60 (2000)
24. Villani, C.: Optimal Transport: Old and New. Springer, Berlin (2009)
25. Weise, T., Wu, Z.: Difficult features of combinatorial optimization problems and algorithm selection with the W-model. In: GECCO Companion, pp. 1453–1460. ACM, New York (2014)